

MODEL OPTIMIZATION REPORT

House Price Prediction Using Machine Learning

Dataset: Ames Housing Dataset

Best Algorithm: CatBoost Regressor

Optimization Techniques: Cross Validation, Hyperparameter Tuning

1. Introduction

1.1 Overview

After comparing multiple machine learning regression models, CatBoost Regressor achieved the best prediction performance. To further improve the model's accuracy and reliability, optimization techniques such as Cross Validation and Hyperparameter Tuning were applied.

These techniques helped reduce overfitting, improve generalization, and produce a more robust prediction model.

2. Objectives

The objectives of model optimization were:

- Improve prediction accuracy.
- Reduce prediction error.
- Validate model stability.
- Find the best hyperparameter combination.
- Build a reliable final prediction model.

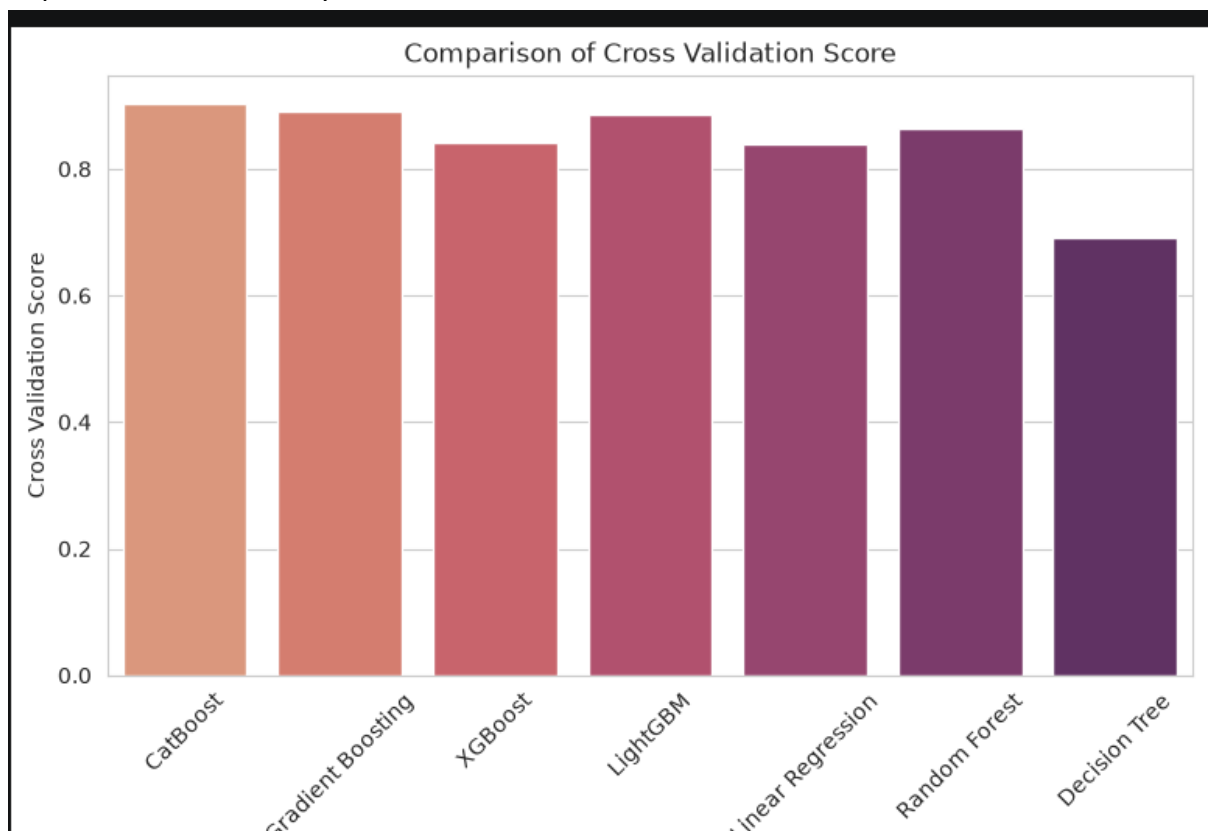
3. Cross Validation

Cross Validation is a resampling technique used to evaluate how well a machine learning model performs on unseen data.

In this project, **K-Fold Cross Validation** was performed on the CatBoost Regressor.

Purpose

- Reduce overfitting.
- Evaluate model consistency.
- Improve model reliability.



Cross Validation Chart

4. Cross Validation Results

Model	R2 Score	CV Score
CatBoost	0.907398	0.903125
Gradient Boosting	0.907057	0.889995
XGBoost	0.900009	0.841779
LightGBM	0.899963	0.886280
Linear Regression	0.889701	0.840067
Random Forest	0.881624	0.863269
Decision Tree	0.782153	0.692387

Observation

The cross-validation scores were consistent across all folds, indicating that the model generalizes well and produces stable predictions on unseen data.

5. Hyperparameter Tuning

Hyperparameter tuning was performed to identify the best parameter combination for the CatBoost model.

The following parameters were optimized:

Hyperparameter	Description
Iterations	Number of boosting iterations
Learning Rate	Controls learning speed
Depth	Maximum depth of trees

Grid Search Cross Validation was used to evaluate multiple parameter combinations and select the best-performing configuration.

6. Best Hyperparameters

Feature	Importance
TotalHouseArea	23.325143
OverallQual	15.507806
GrLivArea	4.486953
YearBuilt	3.999863
Fireplaces	3.757660
RemodelAge	3.025577
OverallCond	2.928561
BsmtFinSF1	2.654715
LotArea	2.020130
GarageCars	1.953769
GarageArea	1.800579
BsmtQual_Gd	1.756604
YearRemodAdd	1.702840
HalfBath	1.594750
LotFrontage	1.548950
CentralAir_Y	1.518024
1stFlrSF	1.456833

Parameter	Best Value
Iterations	300
Learning Rate	0.1
Depth	6

7. Optimized Model Performance

Model	MAE	MSE	RMSE	R2 Score	Adjusted R2	CV Score
CatBoost	0.086152	0.017281	0.131456	0.907398	0.037596	0.903125

Gradient Boosting	0.086001	0.017344	0.131697	0.907057	0.034060	0.889995
XGBoost	0.091521	0.018659	0.136600	0.900009	-0.039196	0.841779
LightGBM	0.088333	0.018668	0.136631	0.899963	-0.039668	0.886280
Linear Regression	0.088336	0.020583	0.143468	0.889701	-0.146324	0.840067
Random Forest	0.097788	0.022090	0.148628	0.881624	-0.230266	0.863269
Decision Tree	0.137658	0.040653	0.201625	0.782153	-1.264056	0.692387

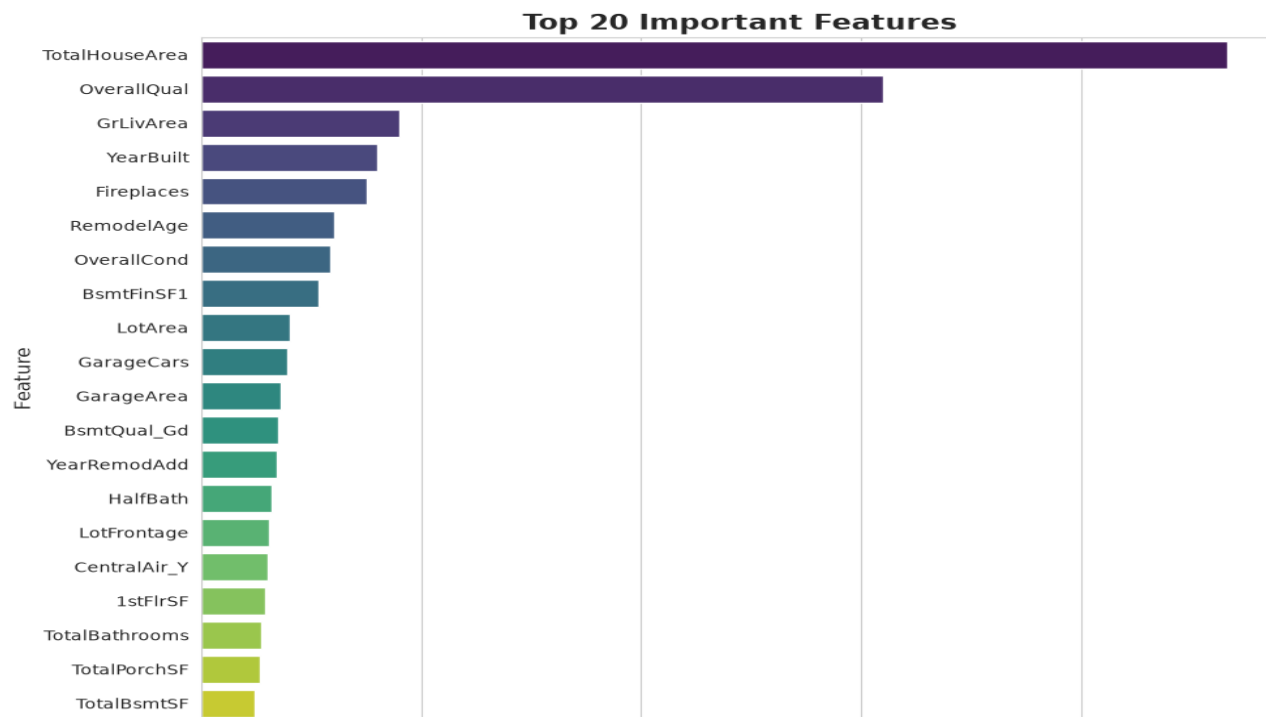
8. Feature Importance

Feature Importance analysis was performed to identify the variables that contributed most to predicting house prices.

The analysis showed that house quality, living area, and engineered features had the greatest influence on the prediction model.

Top Important Features

Rank	Feature
1	TotalHouseArea
2	OverallQual
3	GrLivArea
4	YearBuilt
5	Fireplaces
6	RemodelAge
7	OverallCond
8	BsmtFinSF1
9	LotArea
10	GarageCars



9. Final Model Selection

Based on the optimization results, the CatBoost Regressor was selected as the final prediction model.

Reasons

- Highest prediction accuracy.
- Lowest MAE.
- Lowest RMSE.
- Highest R^2 Score.
- Stable cross-validation performance.
- Strong generalization capability.

10. Summary

Evaluation	Result
Best Algorithm	CatBoost Regressor
Cross Validation	Completed
Hyperparameter Tuning	Completed
Feature Importance	Analyzed
Final Model	Optimized CatBoost

11. Conclusion

Model optimization significantly improved the performance and reliability of the CatBoost Regressor. Cross-validation confirmed that the model maintained consistent accuracy across multiple data splits, while hyperparameter tuning identified the optimal configuration for prediction.

The optimized CatBoost model achieved the best balance between accuracy and generalization, making it the final model selected for deployment in the House Price Prediction Flask application.